

Thien (Theo) Pham

400 N Negley Ave Apt 111, Pittsburgh, PA 15206 | +1.919.448.5730 | quythien14@gmail.com | [linkedin.com/in/quythien](https://www.linkedin.com/in/quythien)

EDUCATION

University of Pittsburgh

Ph.D. Candidate in Biostatistics (GPA: 3.85)

August 2021 – December 2026

University of North Carolina at Chapel Hill

Bachelor of Science in Statistics, Bachelor of Science in Mathematics (Highest Distinction, GPA: 3.85)

Honors: Phi Beta Kappa, Dean's List

September 2017 – May 2021

SKILLS

Programming Languages: R, R Shiny, Python, STATA, SQL, and Bash (proficient); Java, C++, and SAS (familiar).

Data Science: Proficient in Statistical inference, Optimization and Stochastic models, Deep learning (PyTorch), and Machine learning (scikit-learn). Experienced in Data wrangling and Visualization, Bayesian statistics, Clinical trial, and Causal inference.

Research Skills: Developing methods for bio-markers detection in high-dimensional omics data, conducting causal inference and longitudinal analysis for both trial and correlated data, and constructing generative models for the integration of multimodal data.

PROFESSIONAL EXPERIENCE

Otsuka America Pharmaceutical

Senior Statistician Intern

May 2025 – August 2025

- Developed internal SAS pipeline for Bayesian meta-analysis using shared parameter models with advance simulations to validate and support Phase 3 efficacy analysis of siveprelimab in IgA nephropathy.
- Evaluated urinary protein (UP) as a surrogate endpoint for treatment efficacy via mixed-effects and survival modeling.
- Conducted a systematic literature review and applied MNAR multiple-imputation with sensitivity analyses to test the robustness of results from a brexpiprazole + sertraline PTSD trial.

University of Pittsburgh

UPMC - Graduate Research Assistant (Multi-omics Bioinformatics and Bayesian Research Assistant)

August 2021 – Present

- Utilized Hierarchical Bayesian models and Machine learning models (XGBoost, Random Forest, unsupervised clustering) to analyze high-dimensional omics data, facilitating the development of novel concordance metrics for cross-species analysis.
- Evaluated circadian biomarkers from multi-omics data in early-phase oncology trials using high-throughput computing to assess sex- and condition-specific effects across brain regions and cancer tissues in humans and mice.
- Streamlined multi-omics pipelines for preprocessing (STAR, Hisat2), enabling high-fidelity downstream analysis of diverse data types (short-read RNA-seq, single-cell RNA-seq, proteomics, etc.).
- Integrated multi-modal single-cell data using deep learning (Pytorch) and probabilistic models to optimize cellular phenotyping with domain adaptation techniques.
- Applied SmCCNet (phenotype-guided sparse multi-CCA) to combine DNA methylation (450K/EPIC; M-values), RNA-seq, and proteomics with cis-eQTM mapping (± 1 Mb) to derive cross-omics methylation-gene-protein modules.
- Designed and implemented graph-based machine learning models in PyTorch Geometric to infer causal relationships between genes and diseases from biological network data.

Department of Infectious Disease - Graduate Research Assistant

August 2023 – Present

- Applied Mixed-effect models and Generalized Estimating Equations (GEE) to analyze clinical trial data, deriving insights to improve HIV treatment adherence and quality of life, particularly among women workers in industrial zones.
- Implemented longitudinal mediation models to estimate short-term causal effects of dry needling, identifying pain reduction as a significant mediator of disability outcomes.

Carolina Center for Neurostimulation, Chapel Hill, NC

Undergraduate Research Assistant

August 2019 – December 2019

- Investigated cortical synchrony between the thalamus and cortex using Python and MATLAB to enhance models visualizing brain interactions, advancing brain stimulation research, and understanding of cortical dynamics.

Hill-Rom, Cary, NC

Research Assistant Intern

June 2020 – August 2020

- Conducted comprehensive exploratory data analysis and literature review to provide the team with key insights for development of Machine learning models for predicting pediatric sepsis in ICU settings.

Other Projects

Deep learning course projects

- Implemented deep learning models, including CNNs and RNNs, integrating VAEs and efficient sampling techniques to improve phoneme recognition and alignment with spoken inputs.
- Implemented diffusion models on ImageNet-100 with DDIM, VAE, and EMA, achieving better performance on reduced-class subsets while addressing high-class dataset challenges.

Publications

Clinical Epidemiology and Public Health

1. Murillo C, Cerezo-Téllez E, Torres-Lacomba M, **Pham TQ**, Lluch E, Falla D, Vo TT. Unraveling the mechanisms behind the short-term effects of dry needling: new insights from a mediation analysis with repeatedly measured mediators and outcomes. Arch Phys Med Rehabil. 2024;105(12):2269-2276. doi:10.1016/j.apmr.2024.07.016

2. Ha T, **Pham TQ**, Misra S, Su TWH, Ha L, Cunningham SD, Schensul SL. Impact of multilevel interventions on reducing reasons for drinking among men living with HIV. AIDS Care. 2025;37(10):1696-1705. doi:10.1080/09540121.2025.2547030

3. Ha T, **Pham TQ**, Nguyen N, Shrestha R, Schensul SL. Prevalence and determinants of induced abortion among women migrant workers in an industrial zone in Vietnam: a cross-sectional study. Medicine (Baltimore). 2025;104(40):e45005. doi:10.1097/MD.00000000000045005

4. **Pham TQ**, Wu Q, Pan C, Ha L, Bui TC, Sujan MSH, Schensul SL, et al. Enhancing quality of life through tailored interventions for men living with HIV. Qual Life Res. 2026;35:27.

5. Almeida Rojo AL, Barnhardt TR, **Pham TQ**, Heim B, Cai L, Tseng GC, et al. Sleep deprivation engages the orexin/hypocretin system to regulate food reward seeking. Int J Neuropsychopharmacol. 2025;28(8):pyaf047.

Multi-Omics and Translational Psychiatry

1. Gober I, Russell A, Shick T, Vagni V, **Pham TQ**, Carlson J, Kochanek P, Wagner A. Systemic administration of soluble glycoprotein 130 promotes cognitive recovery and decreases proinflammatory chemokine inflammation after controlled cortical impact in mice. J Neurotrauma. 2023; 40(15-16):A68.

2. Gober I, Barton D, Awan N, Fan E, Kumar R, Russell A, **Pham TQ**, et al. Exploring temporal dynamics with aromatization after moderate-to-severe TBI and associations with outcome. J Neurotrauma. 2023;40(15-16):A69.

3. Almeida Rojo AL, Barnhardt TR, **Pham TQ**, Heim B, Cai L, Tseng GC, Huang YH. Sleep deprivation engages the orexin/hypocretin system to regulate food reward seeking. Int J Neuropsychopharmacol. 2025;28(8):pyaf047. doi:10.1093/ijnp/pyaf047

4. Barko K, **Pham TQ**, Seney M, et al. Microglial transcriptional responses to SCVS: brain region- and sex-specific profiles. [In Preparation]Barko, K., **Pham, T. Q.**, Seney, M., et al. "Sex Differences Detected in the Proteome of MDD Subjects." [In Preparation]

5. Barko K, **Pham TQ**, Seney M, et al. Sex differences detected in the proteome of MDD subjects. [Under Review]

6. Shelton M, **Pham TQ**, Seney M, et al. Genetic and gonadal sex transcriptional factors underlying sex differences in stress susceptibility in the prefrontal cortex and nucleus accumbens. [In Preparation]

AWARDS

• Travel Funding Award – STATGEN Conference (2025)	2025
• Biostatistics Dean’s Day Doctoral Award	2025
• Outstanding Teaching Assistant (TA) Award	2025
• Phi Beta Kappa Induction, top 1% , University of North Carolina at Chapel Hill	2021
• International Championship of Collegiate A Cappella – Semi-finalist, rank 2nd	2020
• International Championship of Collegiate A Cappella – Semi-finalist, rank 2nd	2019
• Outstanding student Scholarship, top 1% , Pham Ngoc Thach University of Medicine	2016
• Outstanding student Scholarship, top 1% , Pham Ngoc Thach University of Medicine	2015
• Excellent student at City level in Biology, rank 2nd	2014
• Excellent student at City level in Biology, rank 2nd	2013
• Excellent student at City level in Biology, rank 2nd	2011

TEACHING EXPERIENCE

University of Pittsburgh

Years Taught

Department of Biostatistics, Graduate Teaching Assistant

• BIOST 2086: Mixed Models	Spring 2024
• BIOST 2050: Longitudinal and Clustered Data Analysis	Fall 2023
• BIOST 2041: Introduction to Statistical Methods	Spring 2022
• BIOST 2038: Foundation of Statistical Theory	Fall 2021, Fall 2022

Institute for Clinical Research Education, Graduate Teaching Assistant

- CLRES 2020: Biostatistics
- CLRES 2005: Computational Methods for Clinical Research

Summer 2022, Summer 2024
Summer 2022, Summer 2024

University of North Carolina – Chapel Hill

Department of Mathematics, Undergraduate Teaching Assistant

- Math 547: Linear Algebra

Fall 2019

MEMBERSHIP & LEADERSHIP

- American Statistical Association (Pittsburgh Chapter), *Member*
- Pittsburgh Men's Glee Club, *Member*
- Tar Heel Voices – UNC's co-ed a cappella group, *Fundraising Committee Member*
- Pham Ngoc Thach University of Medicine, School of Medicine, *Class President*

Years
2021-Present
2023-Present
2019-2021
2014-2016